

PAULO MEDINA

pcmedina.avalos@gmail.com | +240 702-3443 | Harpers Ferry, WV | [LinkedIn](#) | [Portfolio](#)

EDUCATION

Master of Science in Geographic Information Sciences (Currently Enrolled – Graduating this Spring)

University of Maryland, College Park, MD

BA in Forest Engineering

Universidad Nacional de Asunción, Facultad de Ciencias Agrarias, San Lorenzo, PY, May 2021

TECHNICAL SKILLS

- **GIS Software:** ArcGIS Pro, QGIS, ArcGIS Online
- **Programming & Automation:** Python, SQL, R
- **Data Management:** PostgreSQL/PostGIS, MySQL, MySQL Workbench, Python, ArcGIS Portal, AWS Deployment, ETL
- **Reporting/Data Visualization:** Power BI, Tableau, Google Looker, Python, HTML/CSS/JavaScript, Mapbox, Leaflet
- **Remote Sensing:** ENVI, Google Earth Engine, Rasterio, Machine Learning for image classification, spatiotemporal analysis
- **Languages:** Fluent in Spanish, English, and Portuguese; Proficient in French

PROFESSIONAL EXPERIENCE

ICF – On-call Geospatial Data Scientist

Rockville, MD | Jan 2024 – Present

(Full-time: Jan 2024 - Mar 2025; On-call: Jul 2025 - Present)

- Manage acquisition and standardization of **national administrative boundaries (Admin0, Admin1, Admin2)** for multi-country Demographic and Health Surveys, ensuring proper topology and alignment with SSIB standardized boundaries
- Maintain and manage **enterprise GIS infrastructure** for the Spatial Data Repository website, ensuring data quality and accessibility for global health research
- Process and validate **GPS datasets** for surveys across multiple countries including **Mali, Angola, DRC, Timor-Leste, and Malawi**, ensuring compliance with spatial data protocols
- Produce **high-quality thematic maps and spatial visualizations** for final survey reports, presentations, and technical briefings
- Conduct **rigorous quality control checks** on geospatial data prior to publication and release
- Develop **training materials** and co-facilitate the Health Data Mapping workshop to strengthen **geospatial capacity-building** for international staff
- Lead **multilateral meetings** with international colleagues in **English, French, and Portuguese**
- Perform **literature reviews and spatial analysis** for public health and environmental research initiatives
- Develop **data collection applications** using CSharp and JavaScript for field survey operations

- Develop **automation scripts** with Python and Microsoft PowerShell to optimize workflows

City of College Park – Urban Forester (Intern)

College Park, MD | May 2025 – Aug 2025

- Designed **Survey123 forms** for citywide urban forestry inventory and field data collection
- Developed **Python automation scripts** for geocoding workflows using Geopy and Nominatim geocoder
- Created **data processing procedures** documented in Jupyter Notebooks for ArcGIS Dashboards
- Ensured **data quality** for city-wide analytical tools including Crash Data Hub built with ArcGIS Experience Builder
- Supported Residential Tree Planting Program through data collection, visualization, and monitoring using Esri platform

Fielddata – GIS & Data Analyst

Asunción, PY | Jul 2022 – Jan 2023

- Designed and maintained **PostgreSQL/PostGIS geodatabases** for forestry and agricultural project data management
- Created **agricultural parcel shapefiles** for crop monitoring and land management applications
- Processed **GPS tractor line data** for precision agriculture operations including herbicide application, fertilization, and soil preparation for Trimble-assisted driving systems
- Developed **interactive dashboards** in Google Data Studio (Looker), Tableau, and Power BI for agricultural data visualization
- Built **web-based mapping platforms** using HTML, CSS, JavaScript, and Leaflet.js for client data access
- Provided **remote sensing services** using Google Earth Engine to assess crop health through NDVI analysis and land classification
- Created **data collection applications** using AppSheet and VBA for field operations

ACADEMIC EXPERIENCE

Teaching Assistant – Universidad Nacional de Asunción

Forest Restoration | Jan 2023 – Apr 2023

- Supported coursework design, exam development, and hands-on field activities for forest restoration practices
- Supervised two teaching assistants and evaluated student performance

PUBLICATIONS

Medina Avalos, P., González, L., Benítez León, E., Villalba, G. (2021). **Comparison of the initial development of four species in forest restoration plots with three different soil preparation systems in the department of Caazapa, Paraguay.** *Investigación Agraria*, 23(2), 86-93. [Link](#)

Amarilla, S., Villalba, J., Soto, C., Yanosky, A., Pérez de Molas, L., Pinazzo, J., Espinola Torres, M., Peralta-Kulik, N., Gonzalez, R., Bordón, P., Amarilla, S., Cáceres, P., Paniagua, P., Medina, P. (2021). **Strategic Plan of the National System of Protected Wild Areas of Paraguay (SINASIP) 2020-2030.** [Link](#)